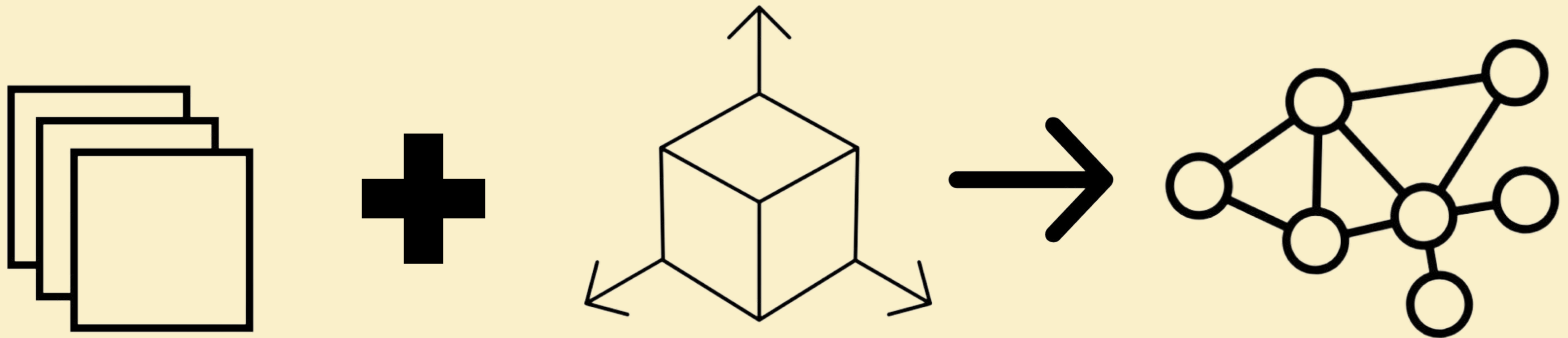


GEOMETRY INFORMED MATERIAL RECOGNITION

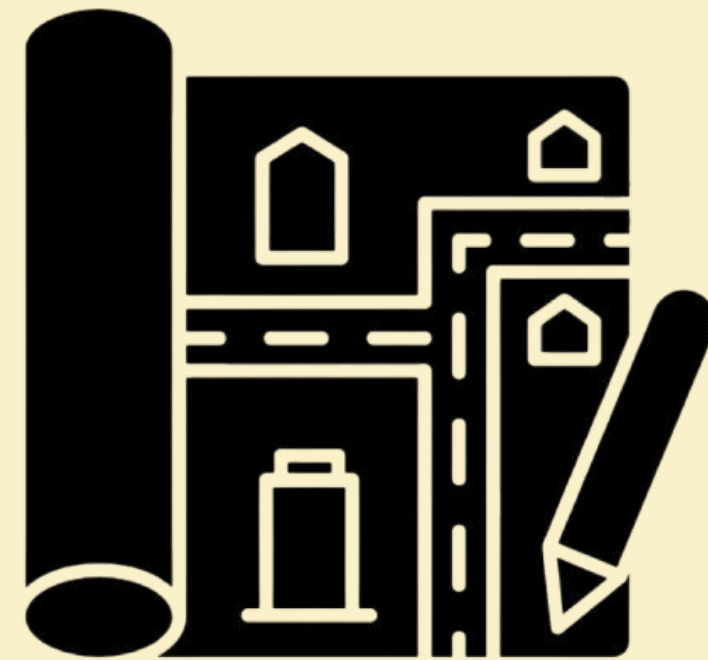
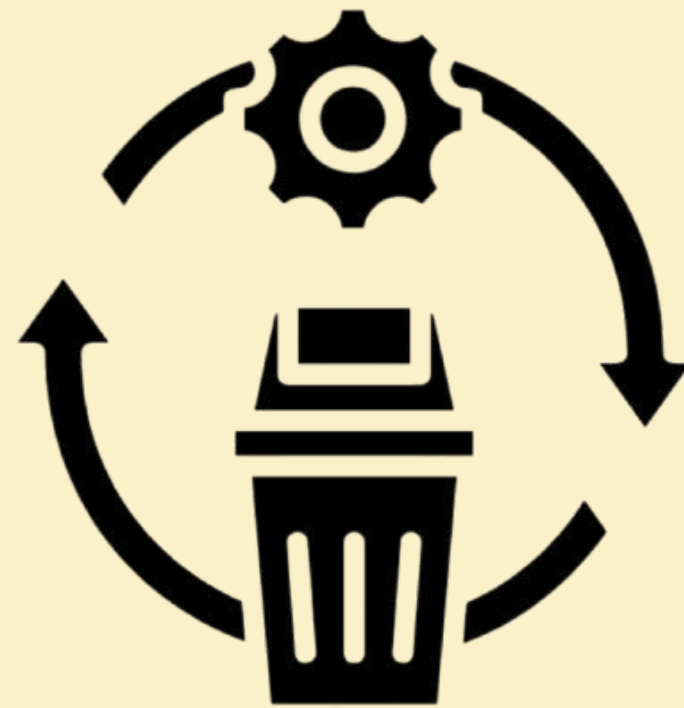
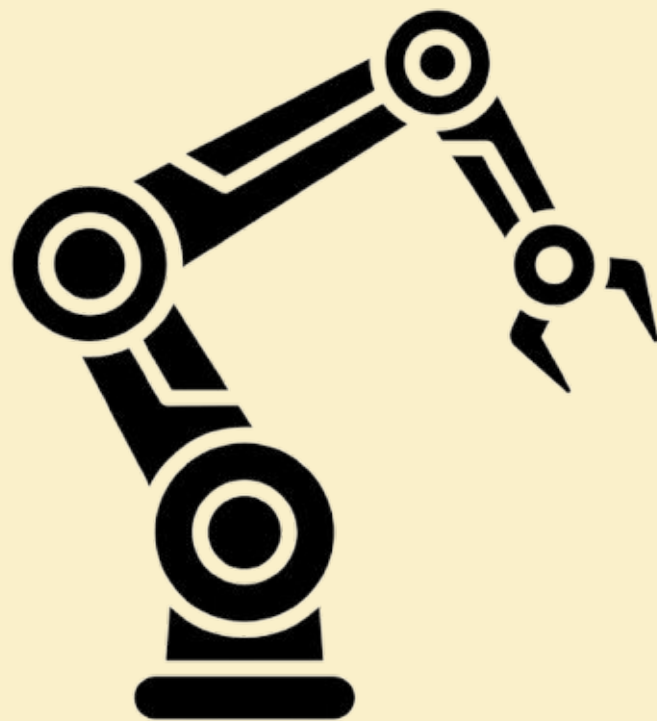
A BRIEF OVERVIEW

This research investigates when geometric data should be introduced to a material classifier and what that suggests about texture-depth correlation.



MATERIAL RECOGNITION

Material recognition allows systems to accurately interface with the real world. Accurate estimations are important for the sorting and separation of materials in various industries including robotics, waste management, agriculture, urban planning, and manufacturing. Enhancing these systems can maximize resource recovery and promote a circular economy.

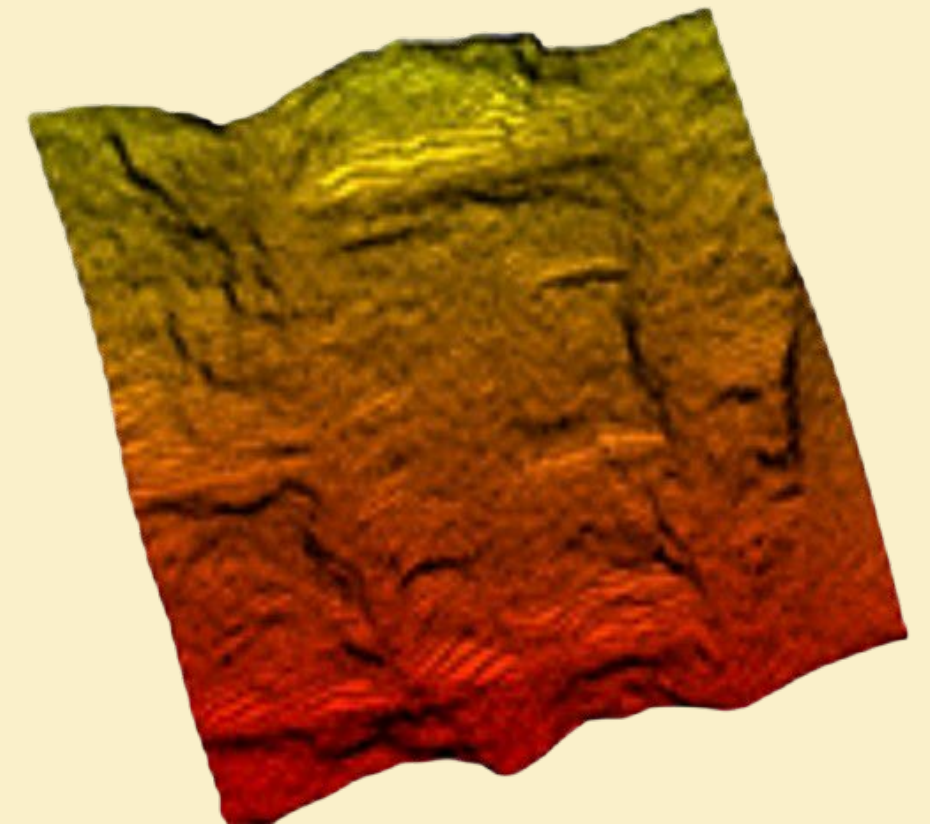
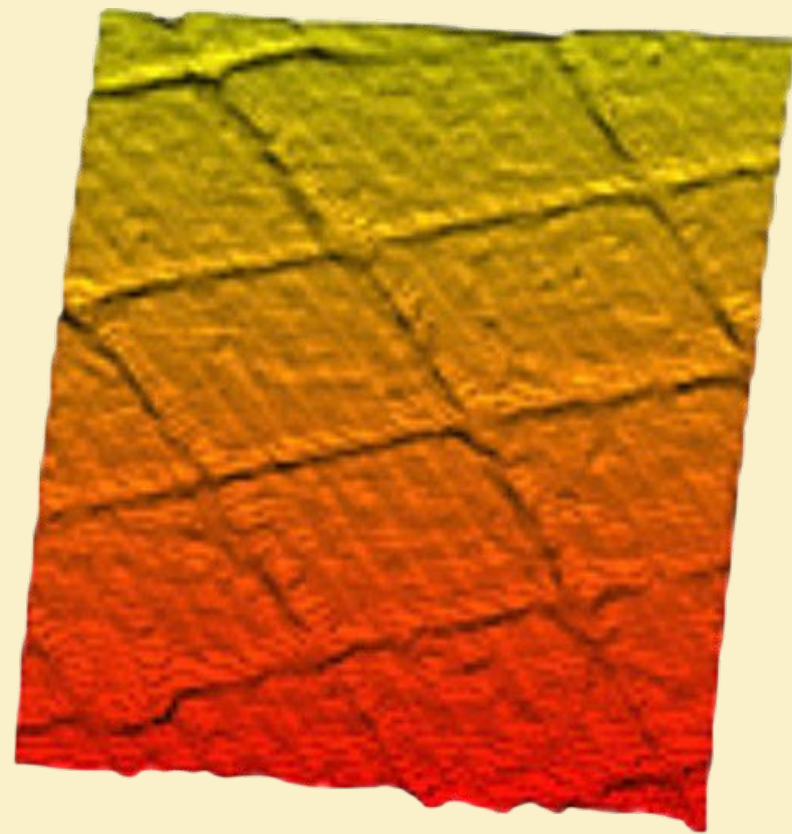
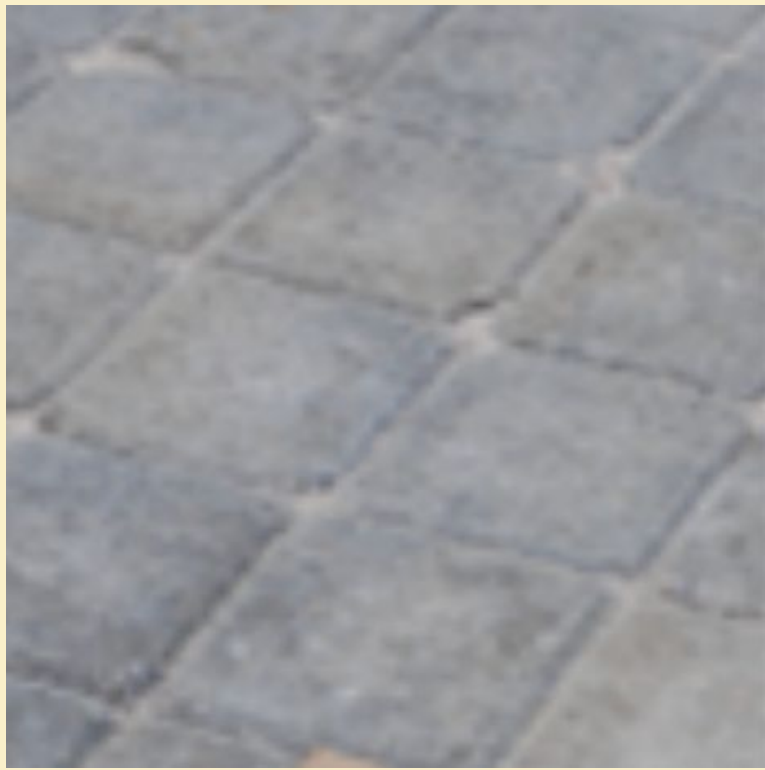


WHAT'S THE PROBLEM?

RGB data alone just does not do enough...

WHY USE DEPTH?

Observed texture is due to a combination of surface markings, micro geometric texture, and camera-relative surface normal. Jointly representing camera-relative surface normals and texture responses may disambiguate similarities between materials.



GEOMAT DATASET

Joseph DeGol, Mani Golparvar-Fard, Derek Hoiem

19
classes

400
training samples per class

100
testing and validation
samples per class

is the Open Access version, provided by the Computer Vision Foundation.
this watermark, it is identical to the version available on IEEE Xplore.

Geometry-Informed Material Recognition

Mani Golparvar-Fard
University of Illinois Urbana-Champaign
{degol2, mgolpar, dhoiem}@illinois.edu

Derek Hoiem

Abstract

material categories using im-
on. In many applications, such
t, coarse geometry information
how 3D geometry (surface nor-
l extrinsic parameters) can be
ture and color) to improve ma-
roduce a new dataset, GeoMat,
e both image and geometry data
and testing patches that were ex-
and perspectives from real world
al category, and (ii) a large scale
at includes 160 images and over
D points. Our results show that
res both jointly and independently
roves classification accuracy across
ewing directions for both material

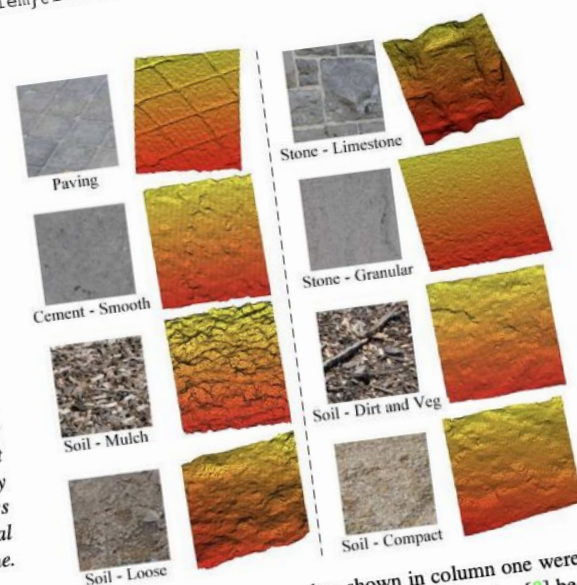


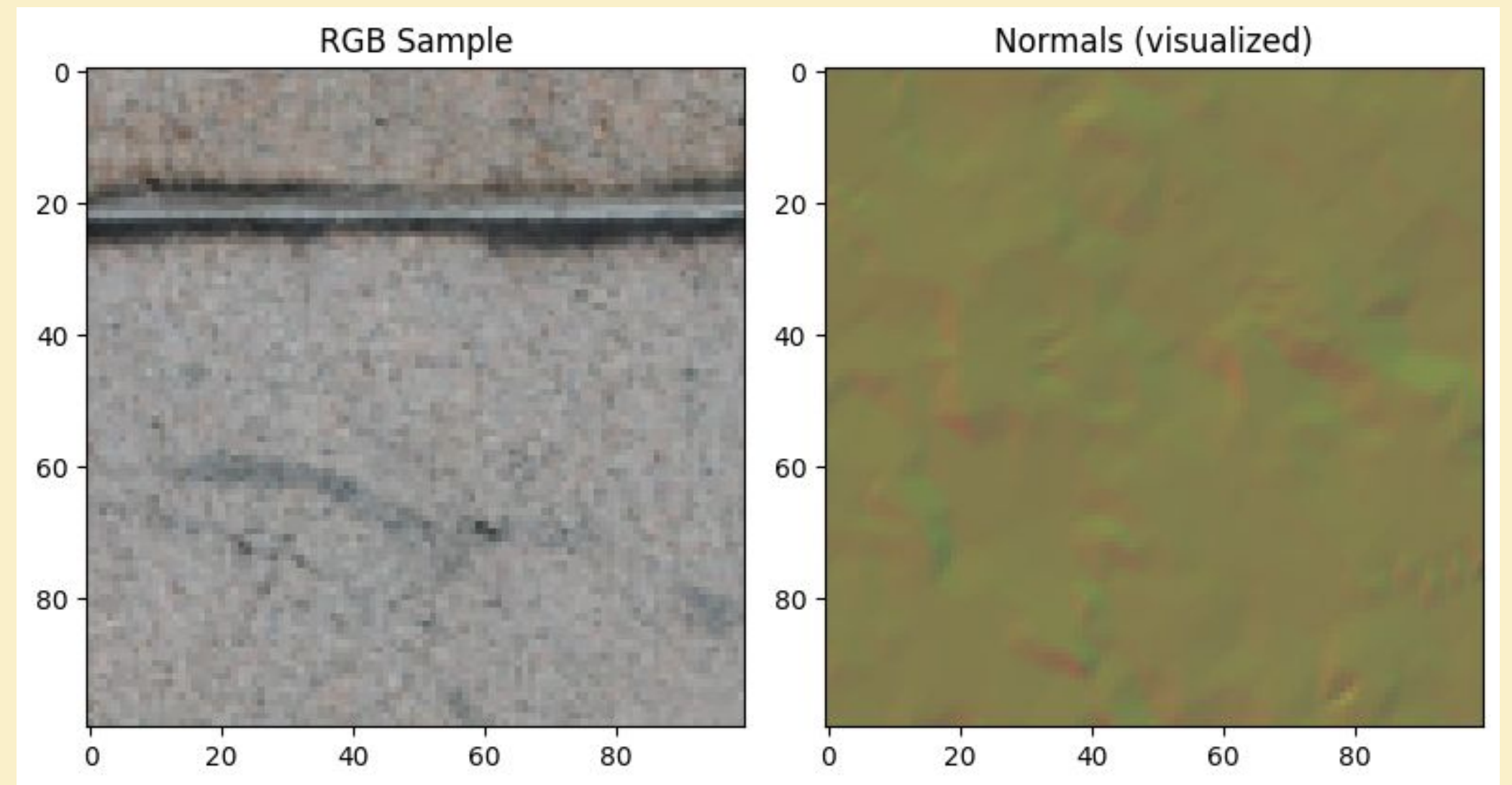
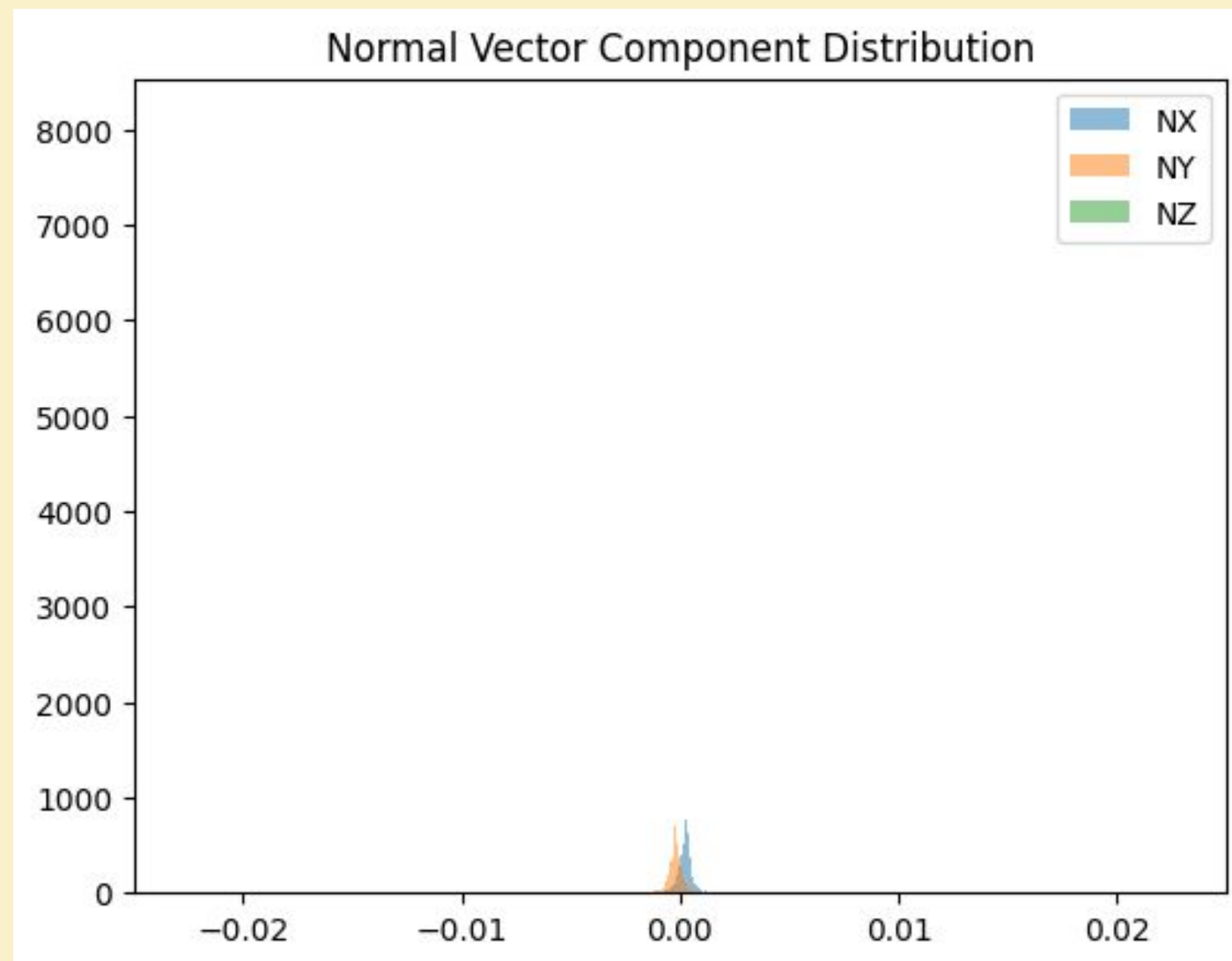
Figure 1: The material patches shown in column one were misclassified as the class shown in column three by [8] because the classes are visually similar. However, the geometry (column two and four) for these patches is different. This paper investigates how to use differences in 3D geometry to improve material classification. We also contribute the GeoMat dataset consisting of images and geometry for material patches and a large scale construction site scene.

gnize material categories using images
ints. In prior material recognition re-
metry is a confounder, and much effort
atures that are stable under varying per-
and rotationally invariant features [33])
ugh the resulting systems often perform
material/texture datasets [11, 16, 5, 25],
not always translate to improved cate-
al objects or scenes [22, 12]. However,
3D surface geometry can
improve

plex ways, it is not clear how best to take advantage of 3D
points. Can local geometry cues be simply added to exist-
texture features, or do they need to be considered
improve robustness of texture

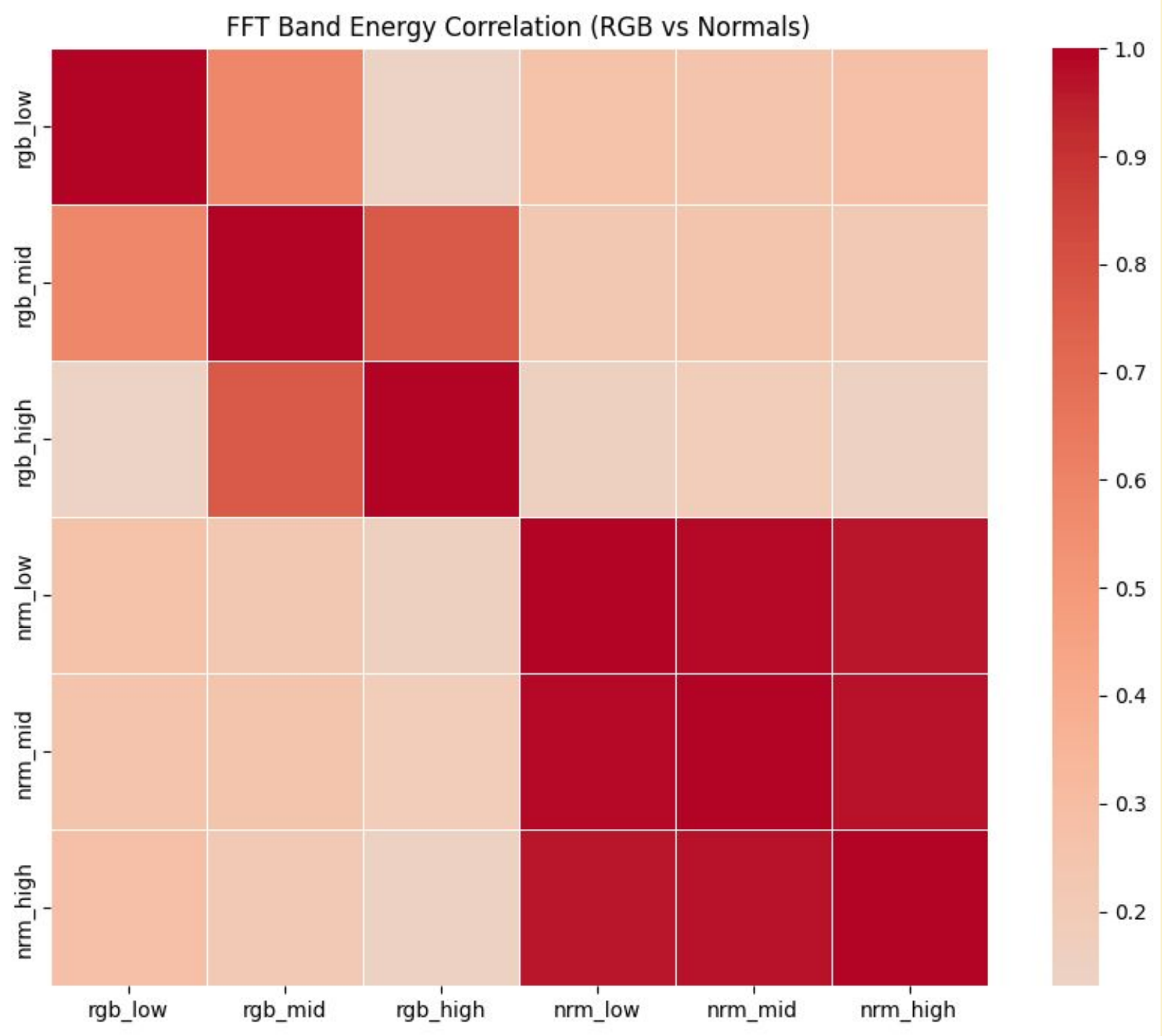


EXPLORATORY DATA ANALYSIS

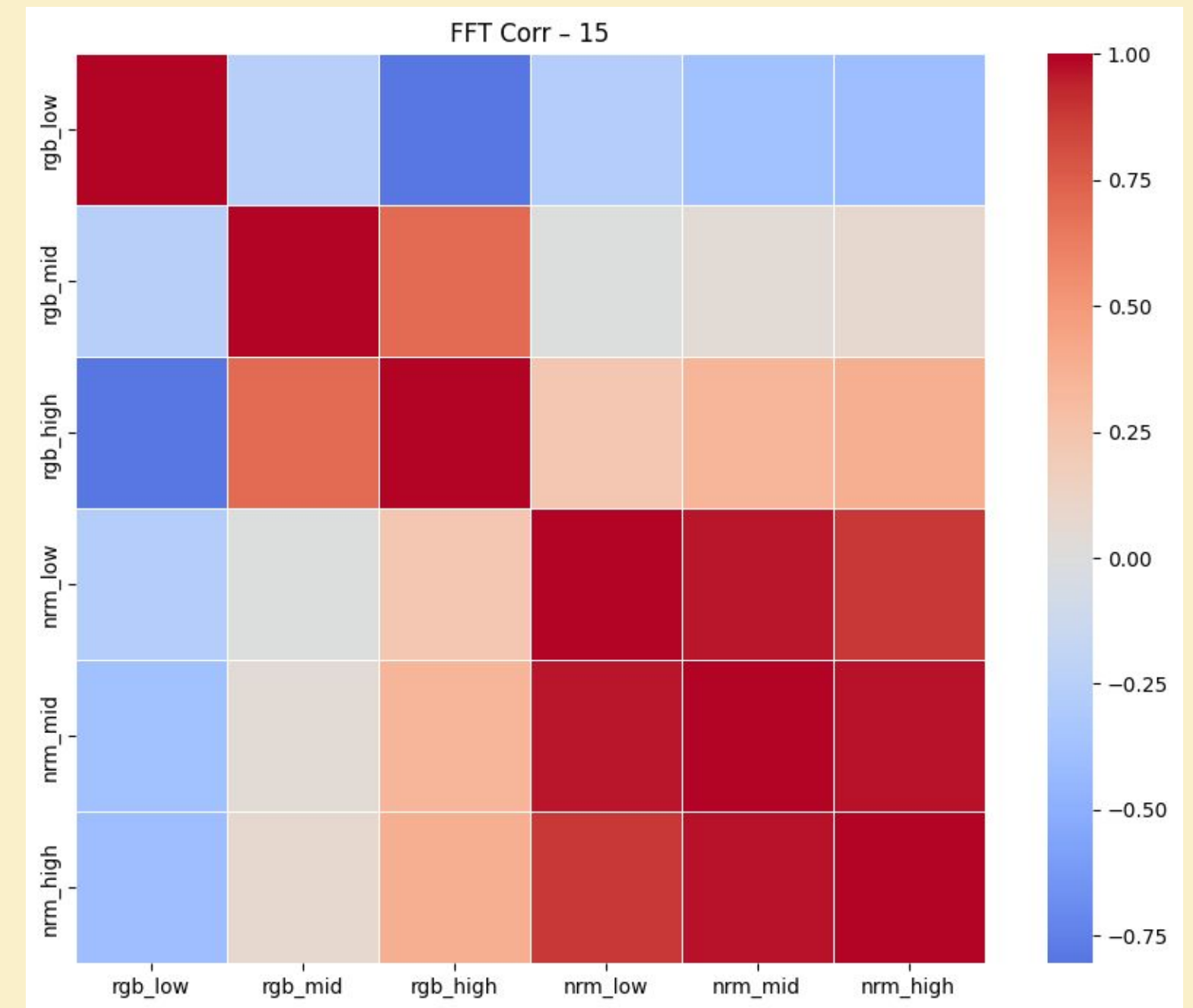
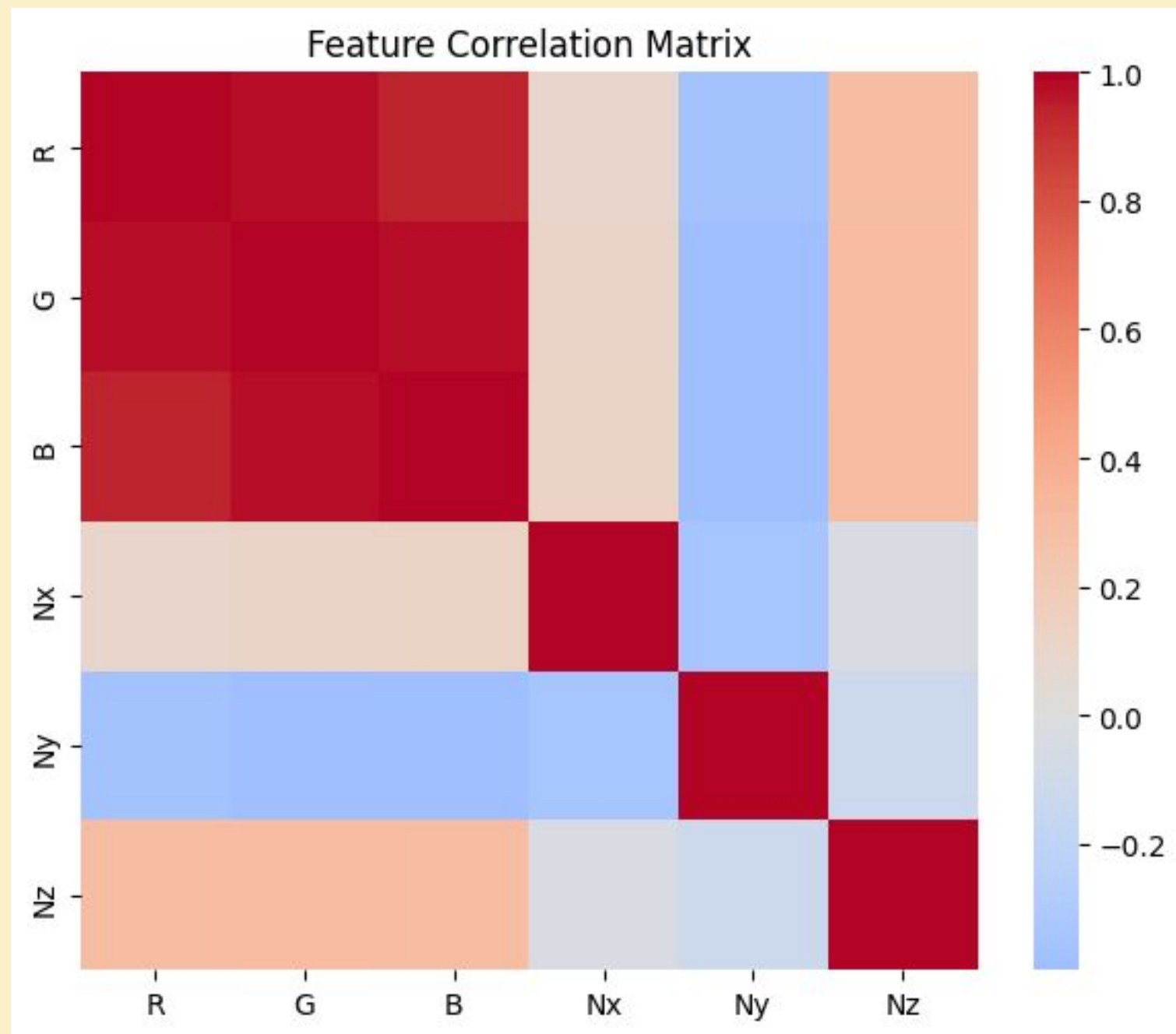


EXPLORATORY DATA ANALYSIS

	nrm_low	nrm_mid	nrm_high	roughness
mean_r	-0.133643	-0.138203	-0.137041	-0.119169
mean_g	-0.042078	-0.050763	-0.047628	-0.040560
mean_b	-0.106950	-0.114689	-0.102997	-0.089832
var_r	0.224454	0.231913	0.229457	0.200056
var_g	0.240156	0.247804	0.246910	0.216888
var_b	0.306762	0.314362	0.312374	0.277558
luma_mean	-0.074724	-0.082904	-0.079208	-0.068286
mean_nz	-0.916602	-0.938869	-0.938187	-0.975952
var_nx	0.900993	0.911973	0.954870	0.989880
var_ny	0.931021	0.946287	0.953288	0.990612
var_nz	0.891621	0.886957	0.884908	0.917983
roughness	0.925987	0.938925	0.963476	1.000000
rgb_low	0.260440	0.256888	0.273748	0.239893
rgb_mid	0.227550	0.243118	0.215605	0.186102
rgb_high	0.157944	0.186341	0.137858	0.116644
nrm_low	1.000000	0.987296	0.961396	0.925987
nrm_mid	0.987296	1.000000	0.972010	0.938925
nrm_high	0.961396	0.972010	1.000000	0.963476



EXPLORATORY DATA ANALYSIS



FEATURE FUSION TESTING

Learning the most optimal fusion can help determine which feature interactions are most meaningful for RGB-D learning, providing a more robust basis for future model attenuations.

Each fusion model learns and leverages the features of the combined data in a different way.

EARLY

MID

LATE

EARLY

input: 6 channels
(R,G,B,Nx,Ny,Nz)

Model: resnet 18
Weights: None

Data combined at input and
processed.

MID

input: 3 RGB channels
input: 3 Normals (Nx, Ny, Nz)

Model: resnet 18
Weights for RGB: "IMAGENET1K_V1"
Weights for Norm: None

Two separate streams,
and then shared head.

LATE

input: 3 RGB channels
input: 3 Normals (Nx, Ny, Nz)

Model: resnet 18
Weights for RGB: "IMAGENET1K_V1"
Weights for Norm: None

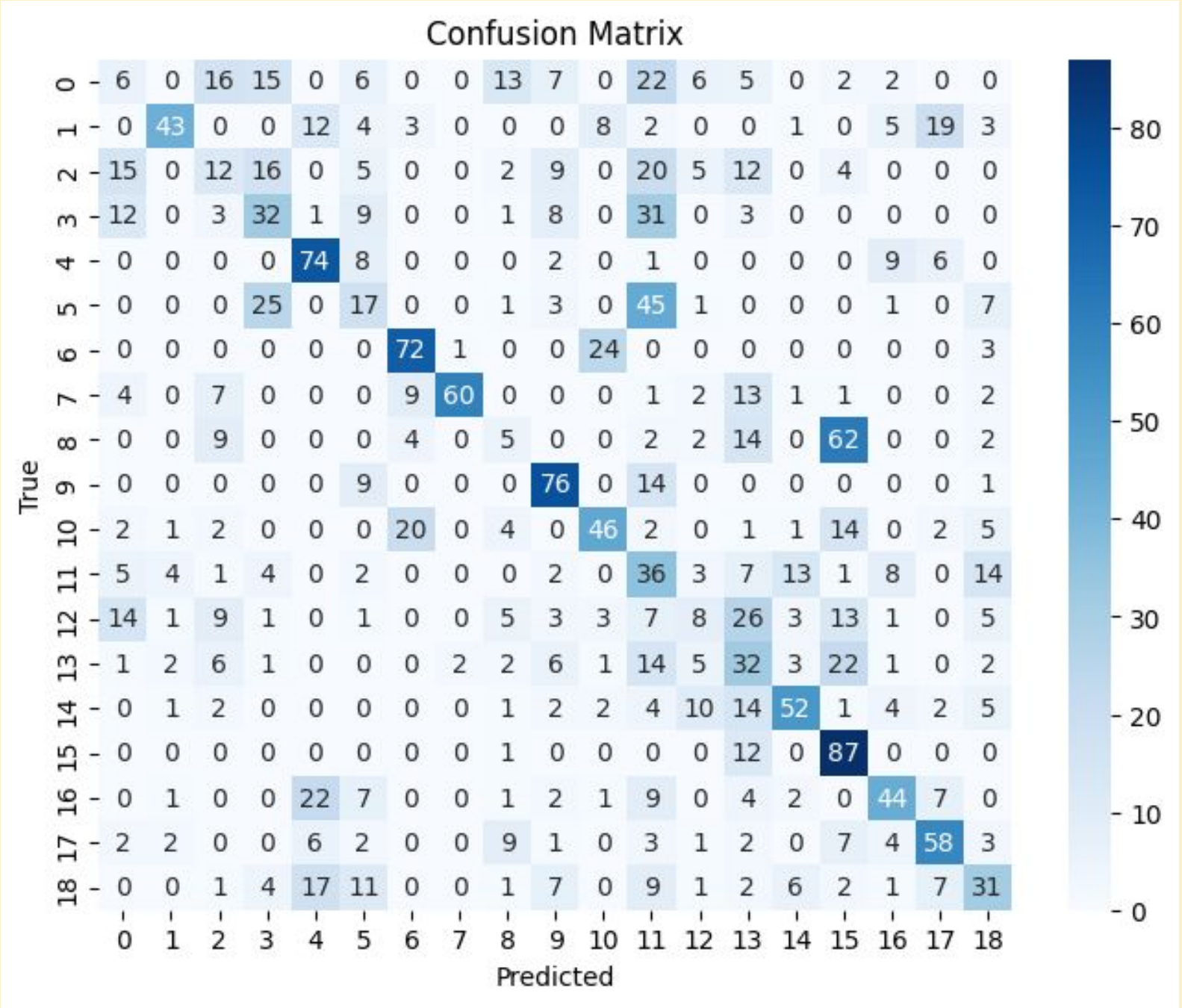
Two separate streams,
concatenated together before output.

All models used ADAM for optimization, Cross Entropy Loss, and a reduce on plateau scheduler.

EARLY FUSION STATS

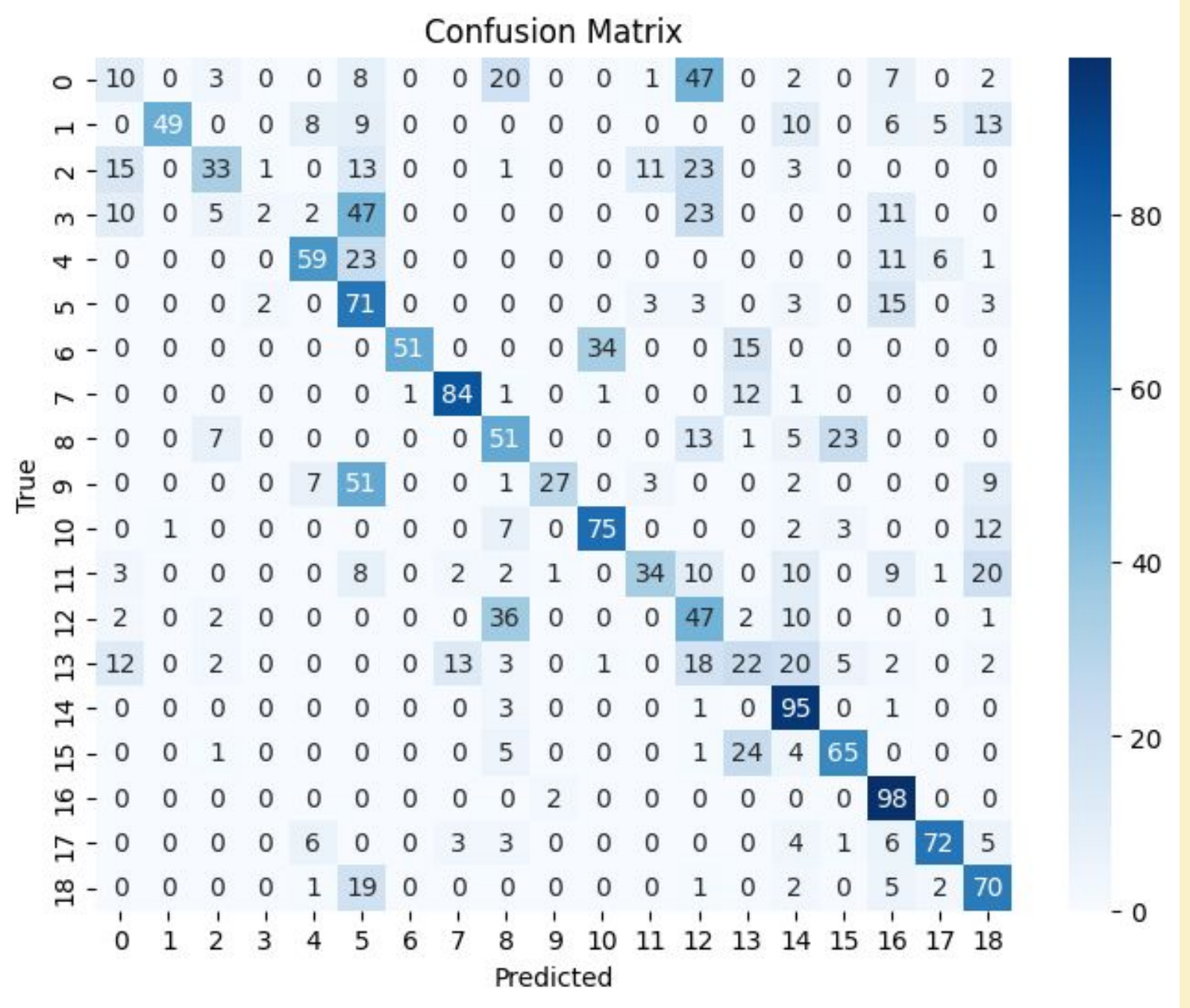
Overall accuracy: 0.4163157894736842

	precision	recall	f1-score	support
Asphalt	0.10	0.06	0.07	100
Brick	0.78	0.43	0.55	100
Cement - Granular	0.18	0.12	0.14	100
Cement - Smooth	0.33	0.32	0.32	100
Concrete - Cast_in_Place	0.56	0.74	0.64	100
Concrete - Precast	0.21	0.17	0.19	100
Foliage	0.67	0.72	0.69	100
Grass	0.95	0.60	0.74	100
Gravel	0.11	0.05	0.07	100
Marble	0.59	0.76	0.67	100
Metal - Grills	0.54	0.46	0.50	100
Paving	0.16	0.36	0.22	100
Soil - Compact	0.18	0.08	0.11	100
Soil - Dirt and Vegetation	0.22	0.32	0.26	100
Soil - Loose	0.63	0.52	0.57	100
Soil - Mulch	0.40	0.87	0.55	100
Stone - Granular	0.55	0.44	0.49	100
Stone - Limestone	0.57	0.58	0.58	100
Wood	0.37	0.31	0.34	100
accuracy			0.42	1900
macro avg	0.43	0.42	0.41	1900
weighted avg	0.43	0.42	0.41	1900



LATE FUSION STATS

	precision	recall	f1-score	support
Asphalt	0.19	0.10	0.13	100
Brick	0.98	0.49	0.65	100
Cement - Granular	0.62	0.33	0.43	100
Cement - Smooth	0.40	0.02	0.04	100
Concrete - Cast_in_Place	0.71	0.59	0.64	100
Concrete - Precast	0.29	0.71	0.41	100
Foliage	0.98	0.51	0.67	100
Grass	0.82	0.84	0.83	100
Gravel	0.38	0.51	0.44	100
Marble	0.90	0.27	0.42	100
Metal - Grills	0.68	0.75	0.71	100
Paving	0.65	0.34	0.45	100
Soil - Compact	0.25	0.47	0.33	100
Soil - Dirt and Vegetation	0.29	0.22	0.25	100
Soil - Loose	0.55	0.95	0.70	100
Soil - Mulch	0.67	0.65	0.66	100
Stone - Granular	0.57	0.98	0.72	100
Stone - Limestone	0.84	0.72	0.77	100
Wood	0.51	0.70	0.59	100
accuracy			0.53	1900
macro avg	0.59	0.53	0.52	1900
weighted avg	0.59	0.53	0.52	1900

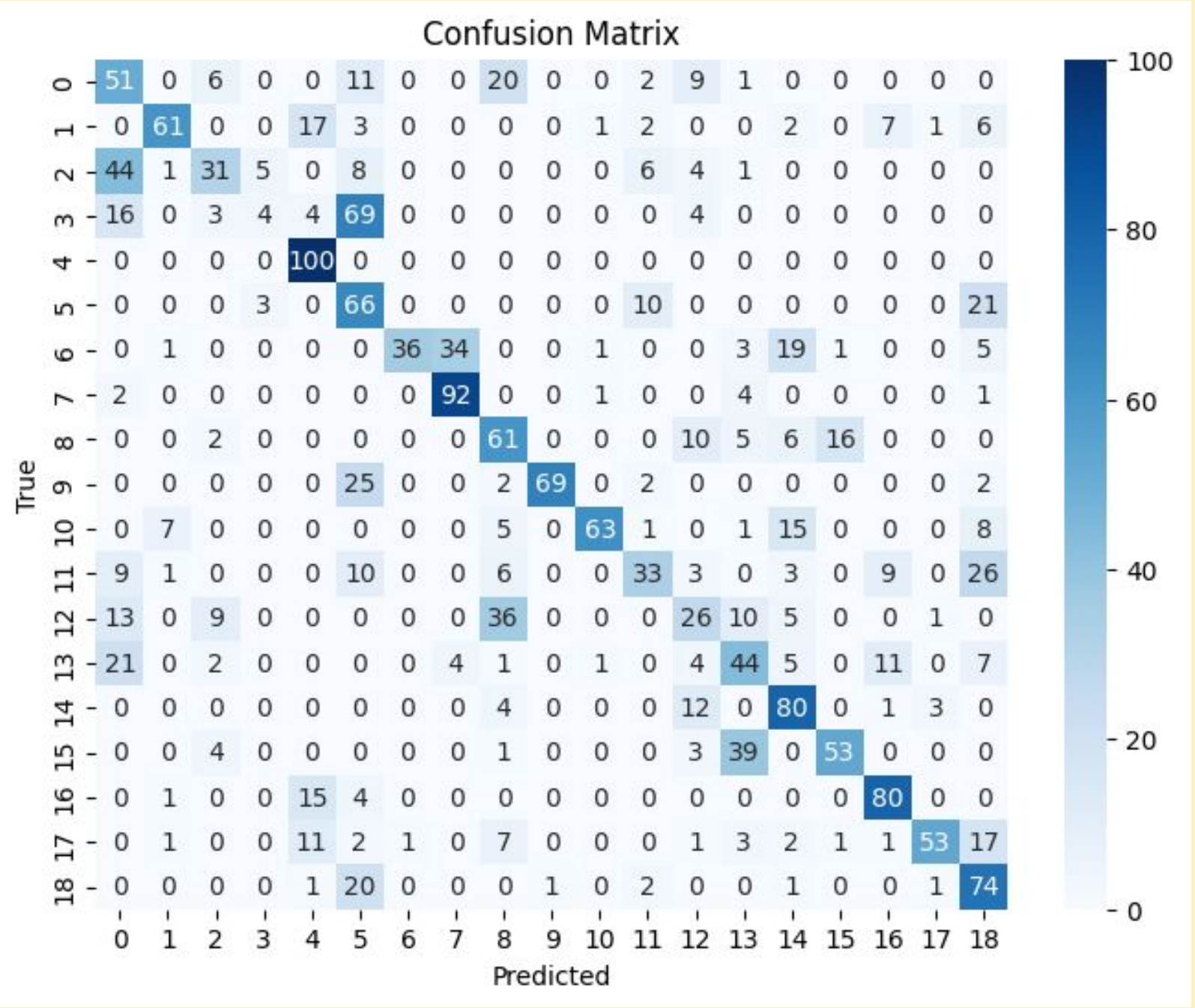


MID FUSION STATS

Overall accuracy: 0.7047368421052631

	precision	recall	f1-score	support
Asphalt	0.72	0.41	0.52	100
Brick	0.74	0.66	0.70	100
Cement - Granular	0.80	0.56	0.66	100
Cement - Smooth	0.62	0.69	0.65	100
Concrete - Cast_in_Place	0.88	0.91	0.89	100
Concrete - Precast	0.54	0.51	0.52	100
Foliage	1.00	0.65	0.79	100
Grass	0.66	0.84	0.74	100
Gravel	0.49	0.65	0.56	100
Marble	0.88	0.96	0.92	100
Metal - Grills	0.97	0.77	0.86	100
Paving	0.65	0.50	0.56	100
Soil - Compact	0.49	0.46	0.48	100
Soil - Dirt and Vegetation	0.44	0.56	0.49	100
Soil - Loose	0.66	0.95	0.78	100
Soil - Mulch	0.84	0.62	0.71	100
Stone - Granular	0.71	0.99	0.83	100
Stone - Limestone	0.80	0.82	0.81	100
Wood	0.82	0.88	0.85	100

accuracy			0.70	1900
macro avg	0.72	0.70	0.70	1900
weighted avg	0.72	0.70	0.70	1900



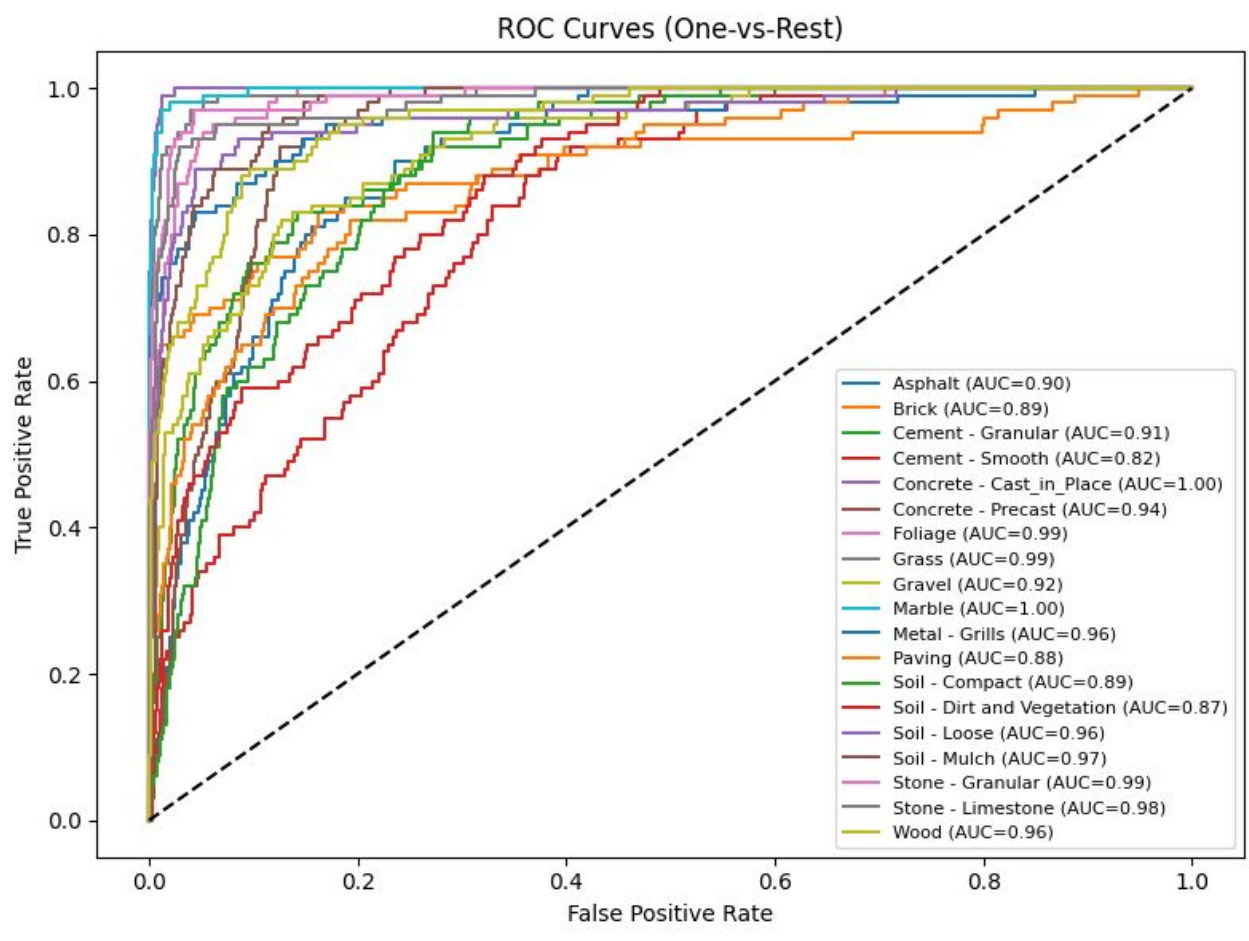
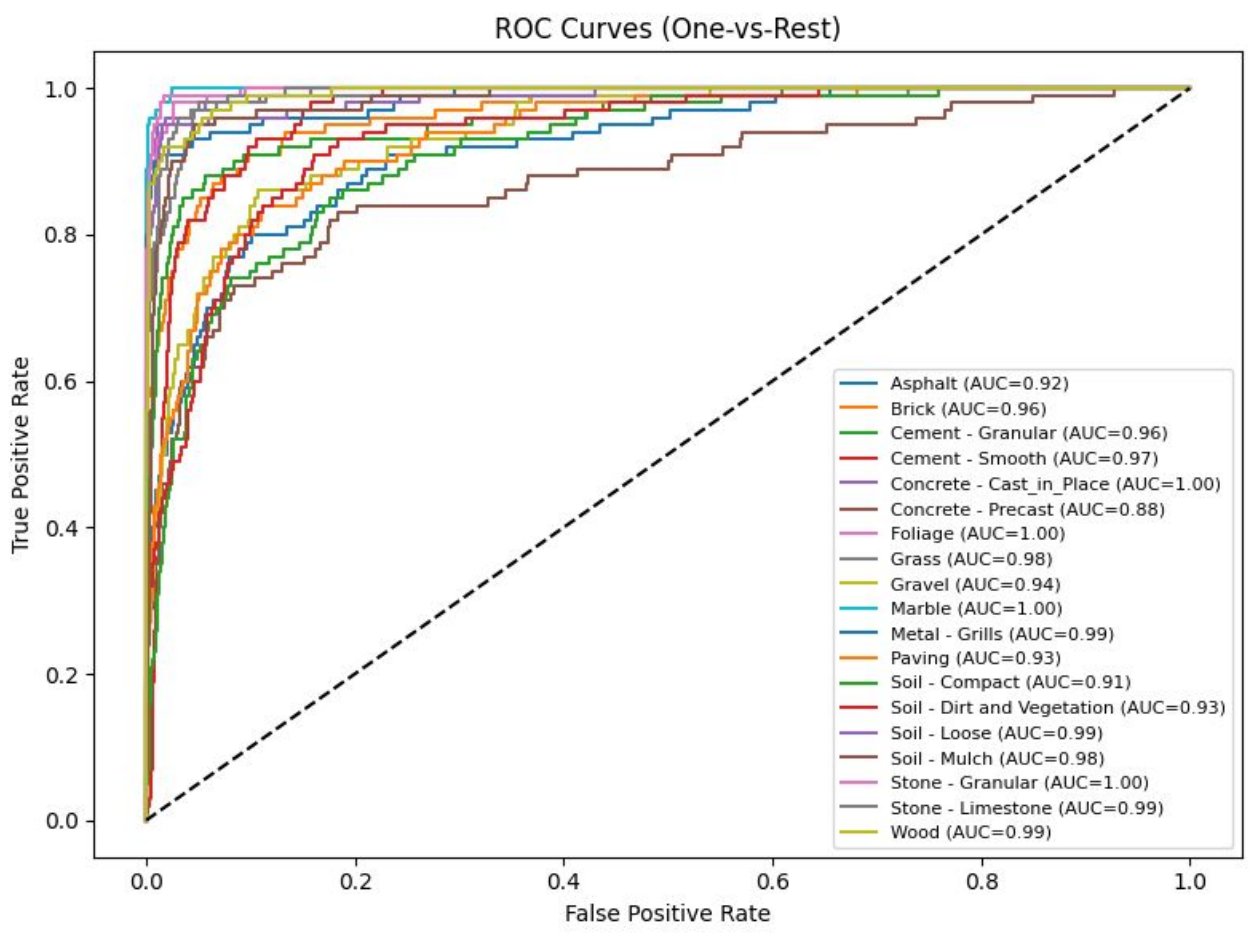
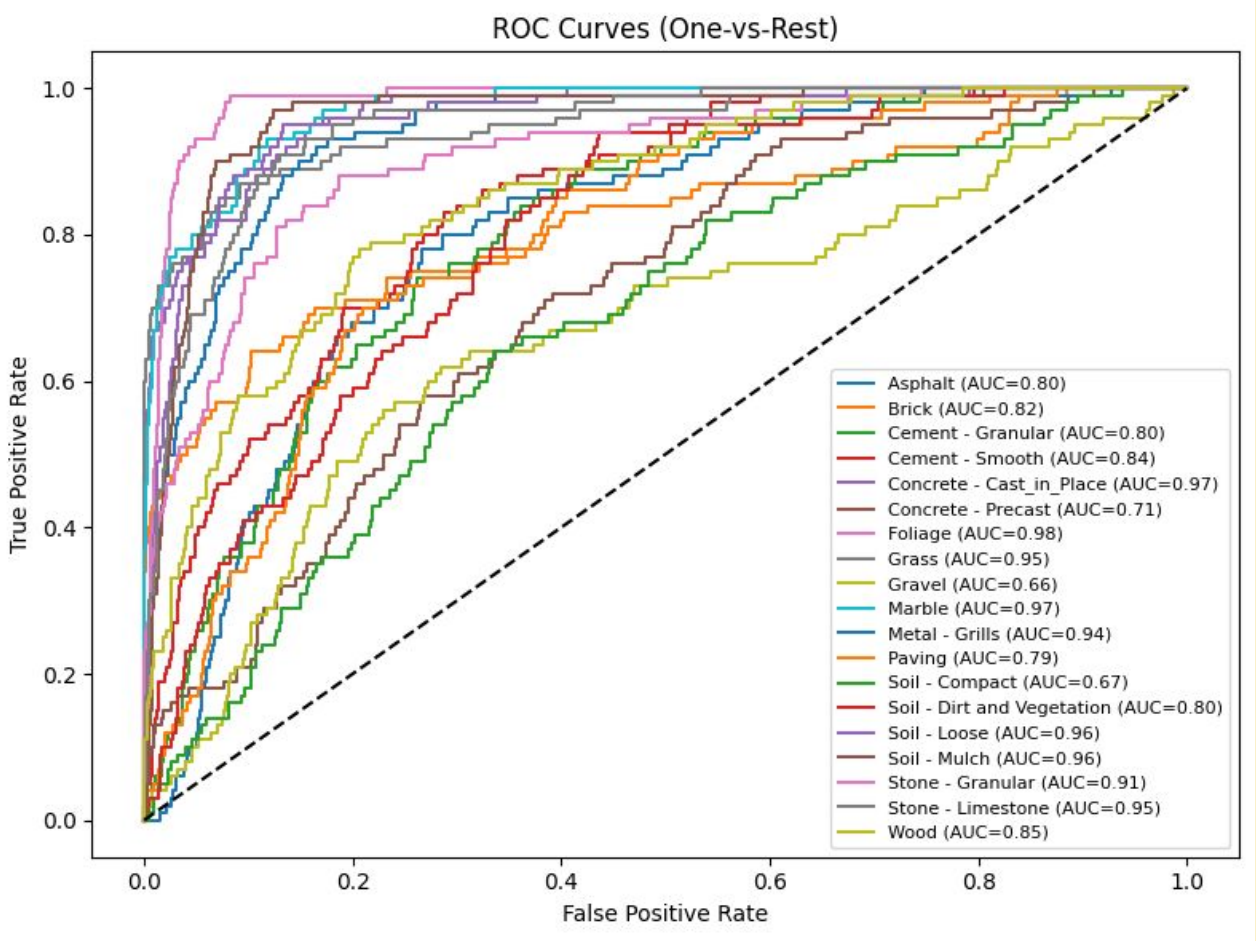
MID

```
Class 0: mean loss = 2.4988
Class 1: mean loss = 2.9804
Class 2: mean loss = 3.1294
Class 3: mean loss = 6.9765
Class 4: mean loss = 0.0109
Class 5: mean loss = 0.9007
Class 6: mean loss = 2.7456
Class 7: mean loss = 0.3142
Class 8: mean loss = 1.8276
Class 9: mean loss = 0.9218
Class 10: mean loss = 2.0260
Class 11: mean loss = 3.2146
Class 12: mean loss = 2.9607
Class 13: mean loss = 2.9040
Class 14: mean loss = 0.9427
Class 15: mean loss = 1.7998
Class 16: mean loss = 0.5646
Class 17: mean loss = 1.7965
Class 18: mean loss = 1.0082
```

LATE

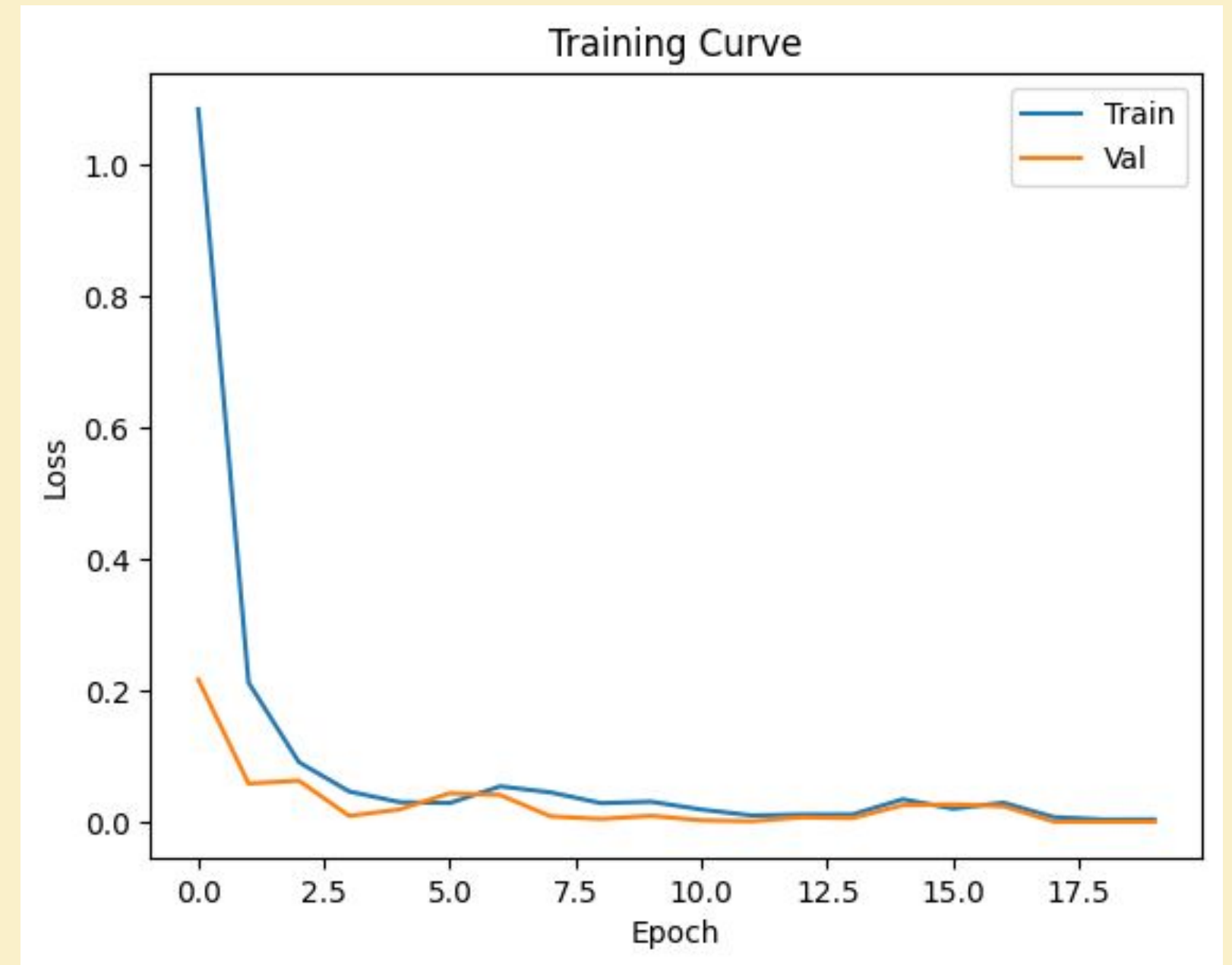
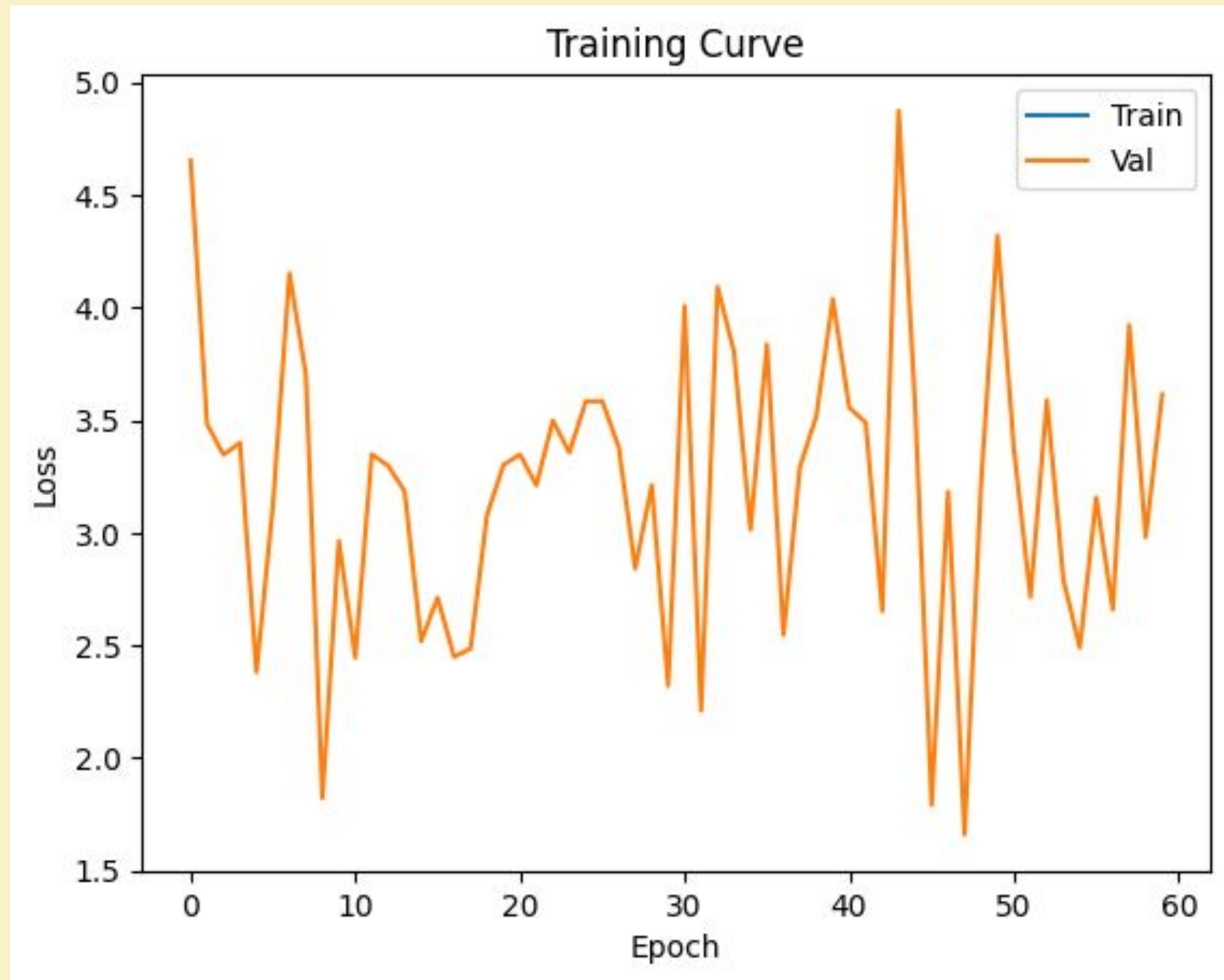
```
Class 0: mean loss = 4.8171
Class 1: mean loss = 3.6723
Class 2: mean loss = 3.2263
Class 3: mean loss = 7.4502
Class 4: mean loss = 1.0517
Class 5: mean loss = 0.9623
Class 6: mean loss = 1.6970
Class 7: mean loss = 0.5102
Class 8: mean loss = 1.6330
Class 9: mean loss = 3.4626
Class 10: mean loss = 0.9188
Class 11: mean loss = 3.9886
Class 12: mean loss = 1.7108
Class 13: mean loss = 3.5220
Class 14: mean loss = 0.2479
Class 15: mean loss = 0.8296
Class 16: mean loss = 0.0484
Class 17: mean loss = 1.0151
Class 18: mean loss = 1.3312
```


ROC (ONE vs REST)





WHAT SUCKED



IMPROVEMENTS

Future research should focus on the mechanism of fusion (e.g., using Attention, Gated Fusion, or specific Tensor operations). This would involve using more advanced techniques where the model learns to adjust the weights of the RGB stream vs. the Normal stream dynamically.

Hyperparameter tuning might improve the classification accuracy of individual models. Tuning learning rate and batch size might benefit the test loss.

Finding datasets with micro-geometry representations